



## Title: Must-Know Maths

### 1) Arithmetic + Algebra

**Know:** Equations, functions, logs/exponentials, powers, manipulating formulas.

**Used for:** Sigmoid/softmax, loss functions, regularisation, interpreting models.

### 2) Basic Probability

**Know:** probability rules, conditional probability, Bayes idea, independence.

**Used for:** Classification probabilities, Naive Bayes, uncertainty, evaluation.

### 3) Statistics

**Know:** Mean/variance/std, distributions (normal idea), correlation, sampling, bias/variance intuition.

**Used for:** Data understanding, noise, overfitting vs underfitting, confidence, metrics.

### 4) Vectors and Matrices (Linear Algebra basics)

**Know:** Vectors, dot product, matrix multiplication, transpose, norms, projections (basic).

**Used for:** Representing datasets as  $X$ , models as  $Xw$ , embeddings, PCA, neural nets.

### 5) Calculus (Enough for gradients)

**Know:** Derivatives, partial derivatives, gradient, chain rule.

**Used for:** Gradient descent, backprop, why updates look like  $X^T(p-y)$ .

That set is enough to understand:

- Linear/logistic regression
- kNN, Naive Bayes (mostly stats/prob)
- Basic neural nets and training
- Metrics and validation properly